year project: 2024/2025

report IUM, python notebooks

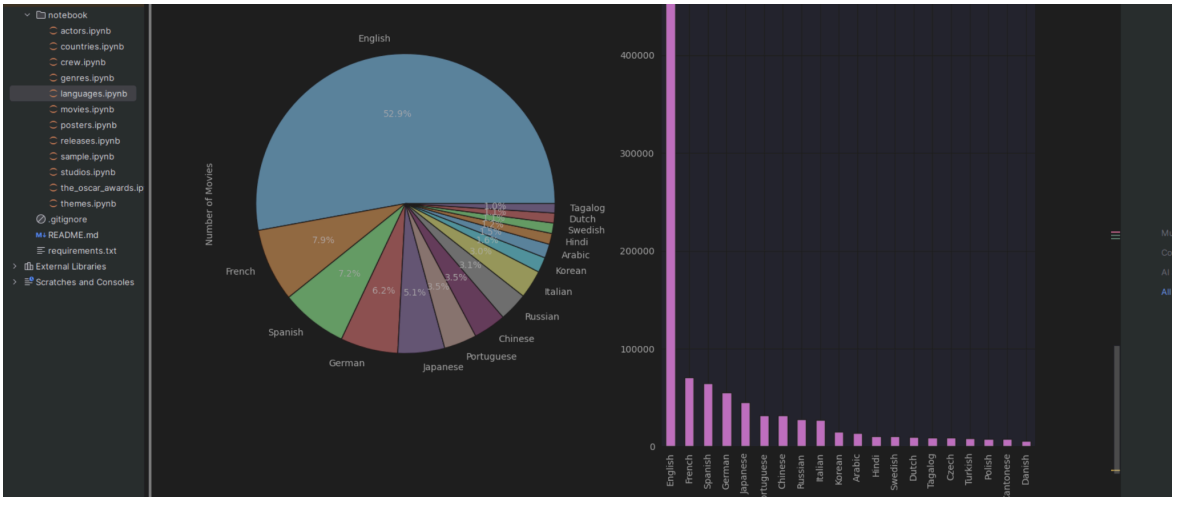
Report IUM

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Introduction

This report aims to explain the work we have done on the python notebooks, In particular we have produced a notebook for each of the csv files given. We have made some analysis, some cleaning and then we produce several plot, using all the libraries seen in class. Then we have created a "sample.ipynb" file where we try to make more complex analysis,



using all the data together, merging different data frames.

Technical tasks

Code Structure:

We organise the project in several notebooks, each of them is provided in a runned format and is autonomous from the others. In each notebook we use a combination of code cells and markdown cells, in order to better explain the analysis on the data we were making.The code cells follow the pattern of producing partial results which will be used in the subsequent ones. All notebooks are stored in a dedicated directory.

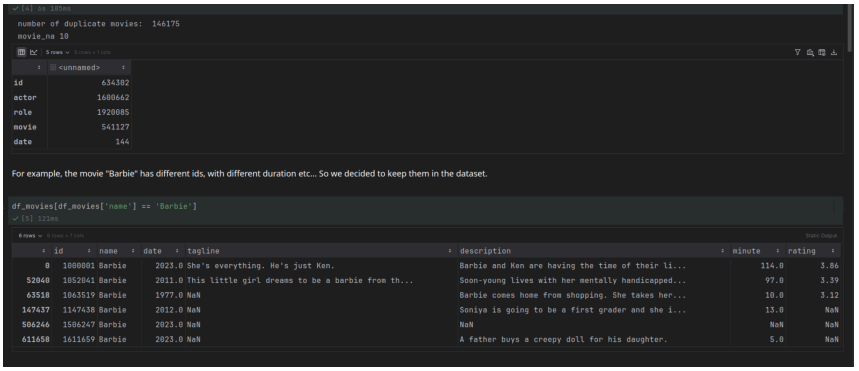
Libraries used:

Pandas was the most used library in the whole project because of its capability to read data from csv, create dataframe, manipulate, visualize and save data; moreover it integrates perfectly with the others libraries we use for data visualization: matplotlib, plotly, seaborn, geopandas.

Data Cleaning:

In a few situations we had to clean some data, and it is reported in the notebook, explained even in

the markdown cell. For Example when we find some movies with duplicated names, there are indeed a lot of them. Another example is when we have to merge the rotten_tomatoes_reviews.csv and the_oscar_awards.csv with the data frames from Letterbox we have to clean e check some of them due the identifiers were not the same, coming from different sites.



Data Analysis and visualization:

We produce several investigations in the notebooks, trying to find hidden correlations among the data. Then showing the results obtained using visualizations with different libraries, as sad before. In particular we use interactive plot and even 3d plot, that are very effective. To show correlations among variables our favourite is seaborn, that offers matrix correlation and heatmap very intelligible.

Database interaction:

We did not directly connect our notebooks to the databases(mongoDB or postgres) as not requested to create a Server in this part of the project. Indeed we read the data from the csv with pandas functions.

Division of work:

We split the workload 50/50

-We developed the notebooks on the single file one each

- About “sample.ipynb” we developed the exact same number of analysis, in particular:

- Bissal Youssef the first four
- Lombardozzi Christian the last four

Conclusion:

We use a large variety of tools to extract, manipulate and visualize data. Trying to use all the technologies seen in class. Anyway it is hard to find correlations between variables except for the obvious ones.